

Research Methods- I (mid- term)

(1)Introduction to Research

1. Meaning of research

The word research is derived from the Middle French "recherché", which means "to go about seeking".

Meaning of research

Research: is simply the process of finding solutions to a problem after a thorough study and analysis of the situational factors.

The term 'Research' consists of two words: Research = Re + Search

'Re' means to observe the phenomena again and again from different dimensions and 'Search' means to find out something.

The word research is derived from the Middle French "recherché", which means "to go about seeking".

Key aspects of research include:

1. **Systematic Inquiry:** Research follows a structured and organized process, involving clear steps and methodologies.
2. **Purposeful Exploration:** Research aims to explore, understand, or explain phenomena, issues, or questions of interest.
3. **Empirical Investigation:** It emphasizes the use of objective and verifiable data to support claims, theories, or findings.
4. **Critical Analysis:** Research involves critical thinking and analysis, where researchers evaluate evidence, identify patterns or trends, and draw conclusions.
5. **Communication of Findings:** Research findings are typically disseminated through scholarly publications, presentations, or reports, allowing other researchers to access, review, and build upon the knowledge generated.

2. Objectives of research

Research objectives are specific goals or aims that guide the conduct of research. They outline the purpose of the study and what it seeks to achieve. Here are some common objectives of research:

1. **Exploration:**
 - **To investigate new phenomena:** This involves examining uncharted areas or new ideas that have not been previously studied.

- **To generate new hypotheses:** Exploration can help identify patterns or trends that suggest new avenues for future research.

2. **Description:**

- **To provide detailed and accurate descriptions:** This involves systematically describing a situation, problem, phenomenon, or group of people to understand it better.
- **To document characteristics:** Research aims to document various aspects such as demographics, behaviors, attitudes, or conditions.

3. **Explanation:**

- **To determine causality:** Researchers seek to identify cause-and-effect relationships between variables.
- **To understand underlying mechanisms:** This involves explaining why and how certain phenomena occur.

4. **Prediction:**

- **To forecast future trends:** Research can aim to predict future occurrences based on current data and trends.
- **To anticipate outcomes:** This involves predicting the outcomes of specific actions or interventions.

5. **Improvement:**

- **To develop better solutions:** Research can focus on improving existing processes, products, or services.
- **To enhance practices:** This involves finding ways to make practices more effective or efficient.

6. **Evaluation:**

- **To assess the effectiveness of interventions:** This involves evaluating the impact of programs, policies, or treatments.
- **To measure outcomes:** Research can aim to measure specific outcomes to determine if objectives have been achieved.

7. **Theory Development:**

- **To construct new theories:** This involves building new theoretical frameworks that can explain phenomena.
- **To refine existing theories:** Research can aim to test and refine existing theories to improve their explanatory power.

8. **Problem-Solving:**

- **To identify solutions to specific problems:** This involves finding ways to address specific issues or challenges.
- **To make informed decisions:** Research can provide the evidence needed to make informed policy or business decisions.

3. Research methods versus research methodology

Research methods and research methodology are two important components of the research process, but they differ in scope and purpose:

1. Research Methods:

- **Definition:** Research methods refer to the specific techniques, procedures, or tools that researchers use to collect, analyze, and interpret data.
- **Examples:** Surveys, experiments, interviews, observations, case studies, content analysis, and statistical analysis are all examples of research methods.
- **Characteristics:**
 - Research methods are concrete and practical approaches employed to gather empirical evidence.
 - They are selected based on the research questions, objectives, and the nature of the data being investigated.
 - Different research methods have different strengths, weaknesses, and suitability for various research contexts.

2. Research Methodology:

- **Definition:** Research methodology refers to the broader framework, principles, and theoretical frameworks that guide the entire research process.
- **Examples:** Research methodology constitutes of research methods, selection criterion of research methods, used in context of research study and explanation of using of a particular method or technique
- **Characteristics:**
 - Research methodology encompasses the theoretical and philosophical perspectives that shape the research approach.
 - It includes decisions about research design, sampling strategy, data collection methods, analysis techniques, and interpretation of findings.
 - Research methodology provides a rationale for selecting specific research methods

4. Types of research in three perspectives.

Application perspective: basic versus applied research.

Basic research and applied research are two fundamental types of scientific inquiry, each serving distinct purposes and goals:

1. Basic Research:

- **Definition:** Basic research, also known as pure or fundamental research, aims to expand our understanding of fundamental principles and phenomena without necessarily having any immediate practical application in mind.
- **Purpose:** The primary goal of basic research is to advance knowledge, uncover new insights, and explore the underlying mechanisms of natural or social phenomena.
- **Characteristics:**
 - Focuses on theoretical concepts and fundamental principles.
 - Often conducted without specific application or practical outcome in mind.
 - Emphasizes curiosity-driven exploration and discovery.
 - Lays the groundwork for further scientific exploration and technological advancements.
- **Examples:** Studies in theoretical physics, fundamental biology, pure mathematics, and philosophical inquiries into the nature of consciousness.

2. Applied Research:

- **Definition:** Applied research aims to solve specific practical problems, address practical concerns, or develop new products, processes, or technologies with direct applications in real-world settings.
- **Purpose:** The primary goal of applied research is to produce practical outcomes, innovations, or solutions that address specific needs or challenges.
- **Characteristics:**
 - Focuses on solving practical problems or meeting specific objectives.
 - Applies existing knowledge and principles to develop practical solutions.
 - Often conducted in collaboration with industry, government, or other stakeholders.
 - Emphasizes the translation of scientific knowledge into tangible benefits for society or industry.
- **Examples:** Medical research to develop new treatments or vaccines, engineering research to design more efficient structures or systems, agricultural research to improve crop yields, and social science research to address societal challenges such as poverty or inequality.

Objective perspective: qualitative versus quantitative research types.

1. Qualitative Research:

- **Nature:** Qualitative research seeks to uncover the complexities, meanings, and underlying reasons behind human behavior, experiences, and interactions.
- **Data Collection:** Qualitative research typically involves collecting non-numerical data through methods such as interviews, focus groups, observations, and textual analysis.
- **Analysis:** Researchers aim to identify patterns, themes, and meanings within the data to generate insights and understanding.
- **Purpose:** It is particularly useful when studying subjective experiences, social interactions, and cultural contexts.

2. Quantitative Research:

- **Nature:** Quantitative research aims to test hypotheses, make predictions, and establish statistical associations between variables.

- **Data Collection:** Quantitative research involves collecting numerical data through structured instruments such as surveys, questionnaires, experiments, or observations.
- **Analysis:** Analysis in quantitative research is statistical and involves techniques such as descriptive statistics, inferential statistics, and correlation analysis.
- **Purpose:** Quantitative research is often used to generalize findings, establish causal relationships, and make predictions about populations.

Mode of enquiry perspective: structured and unstructured approach.

. Structured Approach:

- **Definition:** It emphasizes systematic planning, organization, and clear guidelines for decision-making and implementation.
- **Characteristics:**
 - Clear goals and objectives are established at the outset.
 - The process follows a predetermined sequence of steps or stages.
 - Tasks, roles, and responsibilities are defined and allocated.
 - Data collection, analysis, and decision-making are guided by established methods and criteria.
 - Progress is often monitored and evaluated against predefined benchmarks or metrics.
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2. Unstructured Approach:

- **Definition:** An unstructured approach involves a more flexible, open-ended, and adaptive process without rigid guidelines or predefined steps.
- **Characteristics:**
 - Goals and objectives may be less clearly defined or may evolve over time.
 - The process is more flexible
 - There may be greater emphasis on exploration, experimentation, and innovation.
 - Decision-making may be based on judgment rather than predefined criteria.
 - Progress may be less predictable

Classification of research

Research can be classified into various categories based on different criteria. Here are some common classifications of research:

1. Based on Purpose:

- **Basic Research:** Conducted to expand knowledge and understanding of fundamental principles without immediate application.
- **Applied Research:** Aimed at solving specific practical problems or addressing practical concerns.

2. Based on Methodology:

- **Qualitative Research:** Focuses on exploring and understanding phenomena through non-numerical data collection and analysis methods.
- **Quantitative Research:** Involves collecting and analyzing numerical data to test hypotheses, establish patterns, and make predictions.
- **Mixed-Methods Research:** Combines qualitative and quantitative approaches to provide a more comprehensive understanding of a research problem.

3. Based on Nature of Data:

- **Primary Research:** Involves collecting new data directly from original sources through methods such as surveys, experiments, or observations.
- **Secondary Research:** Involves analyzing existing data or literature to gain insights or answer research questions.

4. Based on Scope:

- **Exploratory Research:** Conducted to explore and gain initial insights into a research problem, often preceding more focused investigations.
- **Descriptive Research:** Aims to describe characteristics or phenomena without attempting to establish causal relationships.
- **Explanatory Research:** Seeks to identify and explain the relationships between variables and establish causality.

5. Based on Time Horizon:

- **Cross-Sectional Research:** Data collected at a single point in time to analyze relationships or differences.
- **Longitudinal Research:** Data collected over an extended period to observe changes or trends over time.

6. Based on Design:

- **Experimental Research:** Involves manipulating variables to observe their effects on outcomes, often conducted in controlled settings.
- **Non-Experimental Research:** Observes and analyzes variables without intervention or manipulation.

7. Based on Domain:

- **Natural Sciences Research:** Investigates natural phenomena and processes, including biology, chemistry, physics, etc.
- **Social Sciences Research:** Focuses on understanding human behavior, society, and interactions, including psychology, sociology, anthropology, etc.
- **Humanities Research:** Explores aspects of human culture, history, language, literature, philosophy, etc.

Inductive versus deductive logic

Inductive and deductive logic are two complementary approaches to reasoning and inference, each with its own principles and methods.

1. Deductive Logic:

- **Definition:** Deductive reasoning starts with a general principle or hypothesis and applies it to specific cases to draw conclusions. It proceeds from the general to the specific.

- **Process:** In deductive reasoning, if the general principle is true then the conclusion must also be true.
- **Example:**
 - Premise 1: All humans are mortal (general principle).
 - Premise 2: Ahmad is a human (specific case).
 - Conclusion: Therefore, Ahmad is mortal.
- **Use:** It is prevalent in scientific hypothesis testing, where researchers derive specific predictions from general theories and then test those predictions empirically.

2. Inductive Logic:

- **Definition:** Inductive reasoning starts with specific observations and seeks to formulate general principles, patterns, or hypotheses based on those observations. It proceeds from the specific to the general.
- **Process:** In inductive reasoning, if the specific observations are true, then there is a probability that the generalization derived from them is also true.
- **Example:**
 - Observation 1: ahmad is mortal
 - Observation 2: ahmad is human
 - Conclusion: Therefore all humans are mortal
- **Use:** It is common in everyday reasoning and decision-making, where people make generalizations based on past experiences or observations.

Time in research: cross sectional versus longitudinal.

In research, the concept of time plays a crucial role in how data is collected and analyzed. Cross-sectional and longitudinal studies are two common approaches used to examine phenomena over time, each offering distinct advantages and limitations.

1. Cross-Sectional Studies:

- **Definition:** Cross-sectional studies collect data at a single point in time, providing a snapshot of a population or phenomenon at that moment.
- **Process:** Researchers gather data from the population of interest at a specific time point. Data collection may involve surveys, interviews, observations, or experiments.
- **Characteristics:**
 - Provides a snapshot of the population at a specific moment.
 - Allows for the examination of relationships between variables at one point in time.
 - Typically less time-consuming and costly compared to longitudinal studies.
 - Useful for identifying patterns, or associations among variables.
- **Example:** A survey conducted to assess the prevalence of depression among college students during a particular semester.
- **Limitations:**
 - Cannot determine temporal (to a particular time) relationships between variables.
 - Does not capture changes or developments over time.
 - Vulnerable where differences between groups may be due to factors other than the variables being studied.

2. Longitudinal Studies:

- **Definition:** Longitudinal studies collect data over an extended period, allowing researchers to track changes and developments over time.
- **Process:** Researchers follow participants over time, collecting data at multiple time points through repeated measurements or assessments.
- **Characteristics:**
 - Allows for the examination of patterns, trends, and changes in variables over time.
 - The identification of temporal sequences of events.
 - Provides insights into individual trajectories and developmental processes.
 - Distinguish between age-related changes and generational differences.
- **Example:** A study following a cohort of individuals from adolescence to adulthood to investigate the long-term effects of early-life experiences on health outcomes.
- **Limitations:**
 - Often requires more time, resources, and participant commitment compared to cross-sectional studies.
 - Participant dropout may affect the validity of results.

(5)Research process

The research process is a systematic method used to gather, analyze, and interpret information to answer a specific question or solve a problem. This process can vary slightly depending on the discipline, but generally, it follows a series of structured steps. Here's an outline of the typical research process:

1. Identifying the Research Problem

Objective: Define the issue or problem you aim to address.

- **Select a Topic:** Choose a broad area of interest.
- **Narrow the Topic:** Focus on a specific aspect to make the research manageable.
- **Formulate the Research Question:** Develop a clear and concise question that guides your investigation.

2. Conducting a Literature Review

Objective: Gather existing information and research related to your topic.

- **Search for Sources:** Use databases, journals, books, and online resources to find relevant literature.
- **Evaluate Sources:** Assess the credibility, relevance, and quality of the sources.
- **Summarize Findings:** Highlight key points, trends, and gaps in the existing research.

3. Developing a Research Hypothesis or Thesis Statement

Objective: State a hypothesis or thesis that your research will test or explore.

- **Hypothesis:** A testable prediction about the relationship between variables.
- **Thesis Statement:** A clear statement of the research purpose or main argument.

4. Designing the Research

Objective: Plan how to collect and analyze data.

- **Choose a Research Method:** Decide on qualitative, quantitative, or mixed methods.
- **Define Variables and Measures:** Specify what you will measure and how.
- **Sampling:** Determine the population and sample size.
- **Data Collection Methods:** Select techniques such as surveys, experiments, interviews, or observations.

5. Collecting Data

Objective: Gather information according to the research design.

- **Execute the Plan:** Follow your data collection methods rigorously.
- **Ensure Accuracy:** Double-check data for consistency and accuracy.
- **Ethical Considerations:** Obtain necessary approvals and ensure ethical standards are maintained.

6. Analyzing Data

Objective: Process and interpret the data to draw conclusions.

- **Data Cleaning:** Remove any errors or inconsistencies.
- **Statistical Analysis:** Use appropriate statistical tools and techniques (for quantitative research).
- **Thematic Analysis:** Identify patterns and themes (for qualitative research).
- **Interpret Results:** Relate findings back to the research question and hypothesis.

7. Interpreting and Discussing Results

Objective: Explain what the data means in the context of the research question.

- **Compare with Literature:** Relate findings to existing studies.
- **Implications:** Discuss the significance of the results for theory, practice, and future research.
- **Limitations:** Acknowledge any limitations or potential biases in the study.

8. Drawing Conclusions and Making Recommendations

Objective: Summarize the findings and suggest next steps.

- **Conclusion:** Provide a clear answer to the research question based on the data.
- **Recommendations:** Suggest practical applications, policy changes, or areas for further research.

9. Reporting and Dissemination

Objective: Share the research findings with a broader audience.

- **Write the Research Report:** Include an introduction, methodology, results, discussion, and conclusion.
- **Presentation:** Prepare to present your findings at conferences or seminars.
- **Publication:** Submit the research to journals, books, or online platforms.

10. Reflecting and Evaluating

Objective: Assess the research process and outcomes.

- **Evaluate the Process:** Reflect on what worked well and what could be improved.
- **Feedback:** Seek feedback from peers or mentors.
- **Continuous Improvement:** Use insights gained to refine future research efforts.

By following these steps, researchers can systematically address questions or problems, ensuring a thorough and credible approach to generating new knowledge.

Importance of research

The importance of research cannot be overstated, as it serves numerous critical functions in society, academia, and various industries. Here are some key reasons why research is essential:

1. **Advancement of Knowledge:** It drives intellectual progress and contributes to the development of various fields, from science and technology to social sciences and humanities.
2. **Problem-solving and Innovation:** Research plays a crucial role in identifying and addressing societal challenges. It drives innovation by uncovering new solutions, technologies, and approaches to complex problems such as healthcare disparities, environmental degradation, and economic inequality.
3. **Evidence-based Decision Making:** It helps policymakers, businesses, and practitioners make informed decisions based on rigorous analysis and evaluation of available evidence.
4. **Improvement of Practices and Interventions:** It identifies best practices, evaluates outcomes, and informs continuous improvement efforts.
5. **Validation and Peer Review:** Research undergoes rigorous peer review (**the system used to assess the quality of a manuscript before it is published**) and validation processes. Peer review helps maintain standards of excellence, fosters accountability, and guards against bias and misinformation.

6. **Education and Training:** It cultivates critical thinking, problem-solving skills, and a spirit of inquiry among students and researchers.
7. **Cultural and Artistic Expression:** Research in the humanities, arts, and cultural studies enriches our understanding of human experiences, creativity, and expression.
8. **Economic Development and Competitiveness:** It creates opportunities for job creation, industry collaboration, and economic diversification.
9. **Health and Well-being:** It contributes to disease prevention, early detection, and the development of new therapies and vaccines.
10. **Global Challenges and Sustainable Development:** Research addresses global challenges such as climate change, biodiversity loss, and poverty alleviation.

(2)Scientific Investigation

Scientific investigation refers to the research methods and techniques that scientists use to develop hypotheses, gather data, conduct experiments, analyze data, and draw conclusions.

Example Observing a cell under a microscope and recording detailed descriptions. Scientific investigations are experimental — for example, treating a cell with a drug while recording changes in the behavior of the cell

The hallmarks of scientific research

The hallmarks are main or distinguishing characteristics of scientific research that ensure its validity, reliability, and effectiveness. Let's explore each of them:

1. **Purposiveness:** Scientific research is purposeful, meaning it is driven by specific objectives, questions, or hypotheses. Researchers embark on studies with clear goals in mind, aiming to address gaps in knowledge, solve problems, or test theories.
2. **Rigor:** Rigor means carefulness and the degree of exactness in research investigation. A good theoretical base and a sound methodological design add **rigor** to a purposive study
2. **Testability:** Scientific hypotheses and theories must be testable, meaning they can be empirically investigated and potentially falsified through observation or experimentation.
3. **Replicability:** Replicability refers to the ability of research findings to be reproduced by other researchers using similar methods and procedures. Replication studies help confirm the reliability and validity of scientific findings and enhance confidence in their generalizability.
4. **Precision and Confidence:** Precision in scientific research refers to the accuracy and exactness of measurements, data, and conclusions. Confidence, on the other hand, reflects the degree of certainty or reliability associated with research findings. Together, precision and confidence ensure that research results are both accurate and trustworthy.

5. Objectivity: The conclusions drawn through the interpretation of the results of data analysis should be objective; that is, they should be based on the facts of the findings derived from actual data, and not on our own subjective or emotional Values

5. **Generalizability:** Generalizability refers to the extent to which research findings can be applied or generalized to broader populations, contexts, or phenomena beyond the specific conditions of the study.

6. Parsimony: Research needs to be conducted in simple and economical manner. Simplicity in explaining the phenomena or problems that occur, and in generating solutions for the problems, is always preferred to complex research frameworks

Economy in research models is achieved when we can build into our research framework a lesser number of variables that explain the variance far more efficiently than a complex set of variables.

The seven-step process in hypothetico-deductive method

The hypothetico-deductive method, popularized by the Austrian philosopher Karl Popper, is a typical version of the scientific method.

The seven-step process in the hypothetico-deductive method is a systematic approach used in scientific inquiry to formulate and test hypotheses. Let's go through each step:

1. **Identification of Broad Problem Area:** The process begins with identifying a broad area of interest or concern within a particular field of study. Researchers explore existing literature, observations, or practical problems to identify an area that warrants further investigation.
2. **Defining Problem Statement:** Once the broad problem area is identified, researchers narrow down the focus by defining a specific problem statement or research question. The problem statement should be clear, concise, and specific, outlining the precise issue or phenomenon to be investigated.
3. **Developing Hypothesis:** Based on the problem statement, researchers develop one or more hypotheses that propose a tentative explanation or prediction about the relationship between variables or the outcome of the study. Hypotheses are formulated based on existing theory, prior research, or logical reasoning.
4. **Determining Measures:** Measures are the items in a research study to which the participant responds. Research measures include survey questions, interview questions, or constructed situations. When constructing interviews and surveys, it is important that the questions directly relate to the research questions.
5. **Data Collection:** With measures identified, researchers collect relevant data to test the hypotheses. Data collection methods may include surveys, experiments, observations, interviews, or archival research, depending on the nature of the study and the variables being investigated.
6. **Data Analysis:** Once data is collected, researchers analyze it using appropriate statistical or qualitative techniques to examine relationships, patterns, or trends in the data. The analysis aims to evaluate the hypotheses and determine whether the data support or refute them.
7. **Interpretation of Data:** Finally, researchers interpret the results of the data analysis in light of the research question and hypotheses. They draw conclusions based on the evidence gathered, discussing implications, limitations, and future directions for research.

Some obstacles to conducting research in economics

Conducting research in economics can face several obstacles, ranging from methodological challenges to practical constraints. Here are some common obstacles researchers may encounter:

1. **Data Availability and Quality:** Economics research often relies on data from various sources, including government agencies, surveys, and private datasets. Limited access to relevant data or data of poor quality can hinder researchers' ability to conduct rigorous analysis and draw meaningful conclusions.
2. **Data Limitations and Bias:** Economic data may suffer from limitations or biases, such as measurement error, sampling bias, or data gaps..
3. **Complexity of Economic Phenomena:** Economic systems are complex and dynamic. Understanding these complexities can be challenging, particularly when attempting predict future trends.
4. **Endogeneity and Causality:** Establishing causality in economic relationships can be difficult due to endogeneity, where variables are simultaneously determined by other variables in the system.
5. **Theoretical and Conceptual Challenges:** Researchers may face challenges in selecting appropriate theoretical frameworks or reconciling conflicting theories in their analysis.
6. **Ethical Considerations:** Economic research, particularly in areas such as behavioral economics, may raise ethical concerns related to human subjects, privacy, or potential harm.
7. **Resource Constraints:** Conducting economic research often requires significant resources, including funding, time, and expertise. Limited resources may constrain the scope of research projects.
8. **Political and Ideological Factors:** Economic research can be influenced by political or ideological considerations, particularly in policy-relevant areas. Researchers may face pressure to align their findings with certain perspectives or agendas, potentially compromising the independence and objectivity of their research.
9. **Publication Bias and Academic Incentives:** Academic publishing norms and incentives may favor certain types of research findings or methodologies, leading to publication bias and selective reporting of results.

(3)Research Questions

A research question is a clear, focused, and specific question that guides a research study. It defines the scope of the study, informs the research design, and sets the direction for data collection and analysis. A well-formulated research question is crucial because it helps to:

1. **Clarify the Purpose:** It clearly states what the researcher aims to investigate or understand.
2. **Guide the Research Design:** It influences the choice of research methodology, data collection methods, and analysis techniques.
3. **Limit the Scope:** It ensures the study remains focused and manageable within the given constraints (e.g., time, resources).

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| 4. Provide Direction: It helps to maintain a clear direction throughout the research process. |
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General and specific research questions,

General research question

A general research question will usually be based around 'why' or 'how' a certain phenomenon is happening.

1. They are more general, more abstract and are not themselves directly answerable because they are too general.
2. They guide our thinking, and are of great value in organizing the research project.
3. They are too broad to satisfy the empirical criterion for research questions.

Example

1. What is the relation between family background factors and the incidence of youth suicide?
2. What is the relationship between school experience factors and the incidence of youth suicide?

Specific research questions

Specific questions are **more focused and require a specific answer**. They are used to gather specific information about a subject

1. They are more specific, detailed and concrete.
2. They are directly answerable because the point directly at the data needed to answer them
3. They direct empirical procedures and they are the questions that are actually answered directly by the data in the research.

Example

What is the cause of increased violence among young adults?

Developing research Questions in different situations

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|------------------------------------|--|
| 1. Scientific Research: | <ul style="list-style-type: none"> • Situation: A biologist is studying the effects of climate change on bird migration patterns. • Research Question: How does climate change influence the timing and routes of bird migration in North America? |
| 2. Social Science Research: | |

- Situation: A sociologist wants to understand the impact of social media on adolescent mental health.
- Research Question: What are the relationships between social media usage patterns and symptoms of anxiety and depression among adolescents?

3. Educational Research:

- Situation: An educator is interested in improving student engagement in mathematics classes.
- Research Question: How do different teaching strategies affect student motivation and performance in high school algebra courses?

4. Business Research:

- Situation: A marketing manager is exploring consumer preferences for eco-friendly packaging.
- Research Question: What factors influence consumer perceptions of and willingness to pay for products with sustainable packaging?

5. Medical Research:

- Situation: A medical researcher is investigating potential treatments for Alzheimer's disease.
- Research Question: What are the effects of a novel drug compound on cognitive function and disease progression in patients with mild to moderate Alzheimer's disease?

6. Engineering Research:

- Situation: An engineer is working on developing more efficient solar panel designs.
- Research Question: How can the efficiency of photovoltaic solar panels be optimized through novel materials and structural configurations?

Role of research questions

Research questions play several crucial roles in the research process:

1. **Guiding the Research Process:** They help researchers define the scope of their study and guide them in determining which data to collect, which methods to use, and how to analyze the results.
2. **Formulating Hypotheses:** Research questions often lead to the formulation of hypotheses, which are specific, testable predictions about the relationships between variables.
3. **Generating Knowledge:** Research questions drive the generation of new knowledge by identifying gaps in existing literature or areas where understanding is limited.
4. **Informing Study Design:** Well-formulated research questions ensure that the study design is appropriate for addressing the research objectives.
5. **Facilitating Communication:** Clear and concise research questions facilitate communication among researchers, peers, and stakeholders.
6. **Evaluating Research Outcomes:** The extent to which the research questions are answered or the hypotheses are supported determines the validity and reliability of the study's findings.

Basic types of questions: exploratory and descriptive.

Exploratory and descriptive questions are two basic types of research questions commonly used in various fields of study. Here's a brief overview of each:

1. Exploratory Questions:

- Purpose: Exploratory questions are used to investigate a topic when little is known or understood about it. They aim to explore new areas, generate hypotheses, or identify potential variables of interest.
- Characteristics: Exploratory questions are open-ended and broad, allowing researchers to gather preliminary information and gain insights into the subject matter.
- Example: "What are the factors contributing to employee turnover in the tech industry?"

2. Descriptive Questions:

- Purpose: Descriptive questions seek to describe or characterize a phenomenon, population, or situation. They aim to provide a comprehensive understanding of the topic by examining its characteristics, patterns, or trends.
- Characteristics: Descriptive questions are specific and focused, aiming to gather detailed information about the variables under investigation. They often involve the use of surveys, observations, or archival data.
- Example: "What percentage of high school students in a particular district report experiencing cyberbullying?"

While exploratory questions are useful for generating hypotheses and exploring new areas of inquiry, descriptive questions are valuable for providing detailed descriptions and understanding the current state of a phenomenon. Depending on the research goals and objectives, researchers may use one or both types of questions in their studies.

How problem can be translated into problem statement?

Translating a problem into a problem statement involves clearly defining the issue at hand in a concise, specific, and actionable manner. Here's a step-by-step guide to help you translate a general problem into an effective problem statement:

Steps to Translate a Problem into a Problem Statement

1. Identify the Problem:

- Understand the core issue or challenge.
- Gather relevant information and data to ensure a comprehensive understanding of the problem.

2. Define the Context:

- Describe the background or setting in which the problem occurs.
- Explain why the problem is important and needs addressing.

3. **Specify the Impact:**

- Identify who is affected by the problem and how.
- Highlight the consequences of not addressing the problem.

4. **Narrow the Scope:**

- Focus on specific aspects of the problem that are manageable and solvable.
- Avoid broad or vague descriptions.

5. **Articulate the Problem Statement:**

- Use clear, concise language to describe the problem.
- Ensure the statement is specific, measurable, and focused on the root cause.

Components of an Effective Problem Statement

1. **Introduction:**

- A brief introduction to the issue, providing context and background information.

2. **Statement of the Problem:**

- A precise description of the problem, highlighting the gap between the current state and the desired state.

3. **Evidence:**

- Data or examples that illustrate the problem's existence and severity.

4. **Impact:**

- Explanation of the problem's consequences on stakeholders and the organization.

5. **Goals:**

- The objectives or outcomes expected from solving the problem.

Example

General Problem: "Our Company's customer satisfaction ratings have been declining."

Problem Statement:

Introduction: In the past year, XYZ Company has experienced a significant decline in customer satisfaction ratings.

Statement of the Problem: Current customer satisfaction ratings have dropped from an average of 85% to 65% over the last 12 months.

Evidence: Monthly surveys indicate that customers are increasingly dissatisfied with the quality of our customer service and the responsiveness of our support team. Specific complaints include long wait times, unresolved issues, and a lack of follow-up.

Impact: This decline in satisfaction is leading to increased customer churn, negative reviews, and a reduction in repeat business, ultimately affecting our revenue and market reputation.

Goals: The objective is to identify and address the root causes of customer dissatisfaction, with a target to restore satisfaction ratings to at least 85% within the next six months.

By following these steps and ensuring your problem statement includes these components, you can effectively translate a general problem into a well-defined, actionable problem statement

(4)The Critical Literature Review

A literature review is a critical analysis of existing research literature on a specific topic or research question. It provides an overview of the current state of knowledge, theories, concepts, methodologies, and findings related to the research topic. Here are key aspects of a literature review:

1. **Purpose:** It helps to identify gaps, inconsistencies, or controversies in the literature that justify the need for further research.
2. **Scope:** A literature review should cover relevant studies, publications, and scholarly works related to the research topic. This includes peer-reviewed journal articles, books, conference proceedings, theses, dissertations, and other credible sources.
3. **Organization:** Literature reviews are typically organized thematically, chronologically, or methodologically. Thematic organization involves grouping studies based on common themes or concepts, while chronological organization arranges studies based on the timeline of publication. Methodological organization categorizes studies based on research methods or approaches.
4. **Synthesis:** A literature review involves synthesizing information from various sources to provide a comprehensive overview of the topic.
5. **Critical Evaluation:** This involves assessing the quality of research methods, the validity of findings, the relevance of theoretical frameworks, and the credibility of sources.
6. **Identification of Gaps:** One of the main goals of a literature review is to identify gaps or areas where further research is needed.

7. **Citation and Attribution:** Proper citation of sources are essential in a literature review to acknowledge the contributions of previous researchers and avoid plagiarism..

How to approach the literature review?

Approaching a literature review involves several key steps to ensure a comprehensive and well-organized analysis of existing research on your topic:

Textbooks

Text book is a book used as a standard work for the study of a particular subject.

Textbooks are a useful source of theory in a specific area. An advantage of textbooks is that they can cover a broad range of topics. Textbooks can cover a topic much more thoroughly than articles can.

Journals

A journal is **a scholarly publication containing articles written by researchers, professors and other experts**. Journals focus on a specific discipline or field of study. Both academic and professional journals are important sources of up-to-date information

Theses

A **thesis** is a type of research paper based on your original research. It is usually submitted as the final step of a master's program or a capstone to a bachelor's degree

PhD theses often contain an exhaustive review of the literature in a specific area. Most PhD theses include several empirical chapters. These chapters often have the same structure and characteristics as academic journal articles.

Conference proceedings

Conference proceedings can be useful in providing the latest research, or research that has not (yet) been published. Conference proceedings are very up to date, and for this reason this information source is quite valuable

Unpublished manuscripts

The APA defines an unpublished manuscript as any information source that is not “officially” released by an individual, publishing house, or other company. Examples of unpublished manuscripts may include papers accepted for publication but still “in press,” data from an unpublished study, letters, manuscripts in preparation, and personal communications (including e-mails). Unpublished manuscripts are often very up to date

Reports

Government departments and corporations commission or carry out a large amount of research. Their published findings provide a useful source of specific market, industry, or company information.

Newspapers

Newspapers provide up-to-date business information. They are a useful source of specific market, industry, or company information. Note that opinions in newspapers are not always unbiased.

The Internet

The amount of information that can be found on the World Wide Web is enormous. You can search for (the details of) books, journals and journal articles, and conference proceedings, as well as for specialized data such as company publications and reports. The number of newspapers, magazines, and journals that is available electronically is enormous.

Evaluating the literature

Evaluating the literature involves critically assessing the quality, relevance, and credibility of the sources

1. **Authorship:** Consider the credibility and expertise of the authors.
2. **Publication Outlet:** Examine the publication outlet where the source appears.
3. **Research Methodology:** Evaluate the research methodology employed in the study. Assess the validity, reliability, and generalizability of the research design, sampling methods, data collection procedures, and analytical techniques used in the study.
4. **Evidence and Data:** Examine the evidence and data presented in the source.
5. **Theoretical Framework:** Evaluate the coherence, comprehensiveness, and applicability of the theoretical perspectives or models used to interpret the findings and draw conclusions.
6. **Bias and Conflicts of Interest:** Be alert to potential biases or conflicts of interest that may influence the author's interpretation or presentation of the research.
7. **Relevance and Contribution:** Assess the relevance and contribution of the source to your research question or topic.
8. **Peer Review and Citations:** Peer-reviewed sources undergo rigorous evaluation by experts in the field, enhancing their credibility and reliability. Citations indicate the influence and impact of the source within the scholarly community.

Documenting the literature review

Documenting the literature review involves organizing and presenting the information gathered from the literature in a clear, structured, and scholarly manner.

1. **Introduction:**
 - Provide an overview of the research topic or question.
 - Explain the purpose and objectives of the literature review.
 - Outline the scope and organization of the review.
2. **Search Strategy:**

	- Describe the methods used to identify and select the literature. - Specify the databases, libraries, and search terms used. - Discuss any inclusion or exclusion criteria applied to the selection process.
3. Organization and Structure:	- Choose an appropriate organizational structure for your literature review (e.g., thematic, chronological, methodological). - Create headings or subheadings to organize the literature into coherent sections based on key themes, concepts, or research approaches.
4. Summary of Literature:	- Summarize the main findings, arguments, methodologies, and theoretical frameworks of each source. - Provide brief descriptions or synopses of the key studies, theories, or concepts discussed in the literature.
5. Critical Analysis:	- Analyze and evaluate the strengths and weaknesses of the literature. - Critically assess the quality, relevance, and credibility of the sources. - Identify patterns, trends, controversies, or gaps in the existing research.
6. Synthesis and Integration:	- Synthesize the information gathered from the literature to develop a narrative or argument. - Integrate the findings, theories, and perspectives from different sources to construct a comprehensive understanding of the research topic.
7. Discussion:	- Discuss the implications of the literature for your research question or topic. - Highlight the contributions, limitations, and significance of the existing research. - Identify areas where further research is needed and propose potential avenues for future investigation.
8. Conclusion:	- Summarize the main findings and conclusions of the literature review. - The significance of the research topic and the need for further inquiry. - Recommendations based on the insights gained from the literature.
9. Citations and References:	Provide complete citations for all sources cited or referenced in the literature review.
10. Revision and Editing:	- Review and revise your literature review to ensure clarity, and accuracy. - Edit for grammar, punctuation, spelling, and formatting errors. - Seek feedback from colleagues to improve the quality of your document

Ethical issues: plagiarism.

The ethical issues of documenting the literature review. Research involves building on the work of others. When you summarize, add to, or challenge the work of others, there are two important pitfalls (**difficulty**) that you have to beware of(1) **Purposely (intentionally)** misrepresenting the work of other authors – that is, their viewpoints, ideas, models, findings,

conclusions, interpretations, and so on. **(2) Plagiarism** – the use of another’s original words, arguments, or ideas as though they were your own, even if this is done in good faith, out of carelessness, or out of ignorance. Both purposely misrepresenting the work of others and plagiarism are considered to be fraud

In today’s information age, copying and pasting information from online sources into your own research paper has become very simple. This may create a temptation to copy (significant) portions of text into your work. Your task is to resist this temptation.

Plagiarism is a type of fraud that is taken very seriously in the academic world, mainly because

1. Using the work of others as if it were your own does not convey much respect for the efforts that other people have put into their work.
2. Plagiarism makes it is difficult for the reader to verify whether your claims about other authors and sources are accurate.

Types of plagiarism

Types of plagiarism wherein the sources are not cited

Some people define plagiarism as the copy or borrow of other person’s work, but the words “copy” and “borrow” could actually conceal the gravity of the offense of plagiarism.

Types of plagiarism wherein the sources are not cited are:

1. **The Poor Disguise.** What the writers do in this type of plagiarism is they change how the source looks like by altering the phrases and keywords, while keeping the significant content.
2. **The Ghost Writer.** In this type of plagiarism, the works of another is copied verbatim, and pass it off as the writer’s own.
3. **The Photocopy.** This is the type of plagiarism in which important parts of a particular work of another person are copied by the writer without changing anything.
4. **“The Potluck Paper”**

The writer tries to disguise plagiarism by copying from several different sources

5. **The Self-Stealer.** In self-stealer, the writer borrows, not from other person’s works, but from his or her prior works. This is a violation of the policies regarding the originality expectation that is adopted by nearly all of the academic sectors.

6. **The Labor of Laziness.** Here, rather than spending so much effort in doing the original work, the writer spends his or her time rephrasing the sentences from other works and making them fit together.

Types of plagiarism in which the sources are cited (but still plagiarized)

1. **The Misinformer** is a type of plagiarism wherein finding the sources are not possible because the information given regarding the source is wrong.

2. In the **Resourceful Writer**, all the sentences from other sources are rephrased, the sources are all properly cited, and the quotations are used correctly.

3. The **Forgotten Footnote**. Though the author's name is mentioned as a source, the particular details about the location of the source are disregarded. By hiding the locations of the source, the plagiarist is trying to cover up the many forms of plagiarism.

4. **The Perfect Crime**. The writer gives sources to only some of the information, and rephrases the other sentences without sources. The writer presents the rephrased words as his or her own evaluation.

You can avoid plagiarism by:

1. Keeping track of the **sources** you consult in your research
 2. **Paraphrasing or quoting** from your sources (and adding your own ideas)
 3. Crediting the original author in-text **citation** and in your reference list
 4. Using a **plagiarism checker** before you submit
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